

DeepThinkingMap: Collaborative Video Reflection System with Graph-based Summarizing and Commenting

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ABSTRACT

As people make decisions informed by online videos with varying credibility and quality, reflection on these videos has become increasingly crucial, especially in high-stakes domains such as health-care and finance. However, reflection on videos is challenging for the general public due to the significant cognitive efforts, resources, and skills required. To address this, we present DeepThinkingMap, supporting collaborative reflection and sharing for groups. By facilitating note creation and connection with links, enabling spatial organization, and links between notes and video timestamps, it affords dynamic reflection documenting and sharing experiences for groups. DeepThinkingMap may foster deeper engagement and understanding of online videos, encouraging individuals and groups to reflect critically on the information they encounter.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interactive systems and tools*.

KEYWORDS

Reflection tool, Nudging, Concept Mapping

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1 INTRODUCTION

The unprecedented access to extensive online videos necessitates reflective consumption by individuals and communities, given the varied credibility of user-generated content. This is particularly crucial in high-stakes domains such as healthcare, finance investment, and politics [6, 10]. The research focus and social discussion have been vastly growing about individual and collaborative efforts to mitigate the harm of misinformation and disinformation through content verification, debunking, and moderation [2]. However, factors such as filter bubbles (i.e., the echo chamber where the

perspectives received are confined, shielded from diverse ideas and alternative viewpoints), anchoring bias (i.e., the cognitive tendency to be swayed and influenced by reference points), and cultural and political influences can hinder users from fully reflecting on these resources [2, 3]. Overcoming inertial thinking and engaging in reflective operation requires considerable cognitive effort, which may not be universally accessible or readily available to all individuals and in all situations. Social reflection support, such as synchronous and asynchronous collaborative reflection simultaneously with video consumption, is necessary to encourage people to think beyond literal content [11]. More specifically, it can enhance engagement in reflection and critical analysis when consuming online content, a vital precursor and indicator for further reflective actions.

In this demonstration, we introduce DeepThinkingMap, a graph-based interface designed to facilitate collaborative sharing and reflection during video consumption (see Figure 1). It affords users to identify noteworthy points appearing in the videos, share notes and comments, and, more importantly, connect them in semantic or relational linkage collaboratively. DeepThinkingMap is inspired by other concept mapping and social nudging designs proposed previously [4, 5]. It emerges from the observation that other users' understanding and commenting in collaborative reflection activities may work as nudges for personal reflection. Observing the reactions of other online users to the same videos can yield behavioral insights, such as their level of content engagement. It can also stimulate cognitive thinking based on their responses (e.g., how others react?). These factors could, in turn, foster a deeper personal engagement and reflection [9].

Also, concept mapping as a strategy for learning assessment and guidance, i.e., identifying important ideas and connecting interrelated concepts, is shown to promote inference-making and critical-thinking processes, and can help mitigate the acquisition and accumulation of irrelevant concepts [4, 7]. DeepThinkingMap enables the collaborative video summary and reflection with a more free-format documentation approach.

2 SYSTEM DESIGN AND IMPLEMENTATION

DeepThinkingMap interface amalgamates a graph-based canvas with a video player for collaborative reflection (Figure 1). In the current prototype, users may reflect upon health-related videos preloaded in the system. The standard YouTube video player embedded within the interface provides all typical functions, such as play, pause, and speed adjustments. It accommodates multiple videos per collaboration session, with easy toggling between them. Each user maintains full control of their video player, enabling reflection at their own pace.



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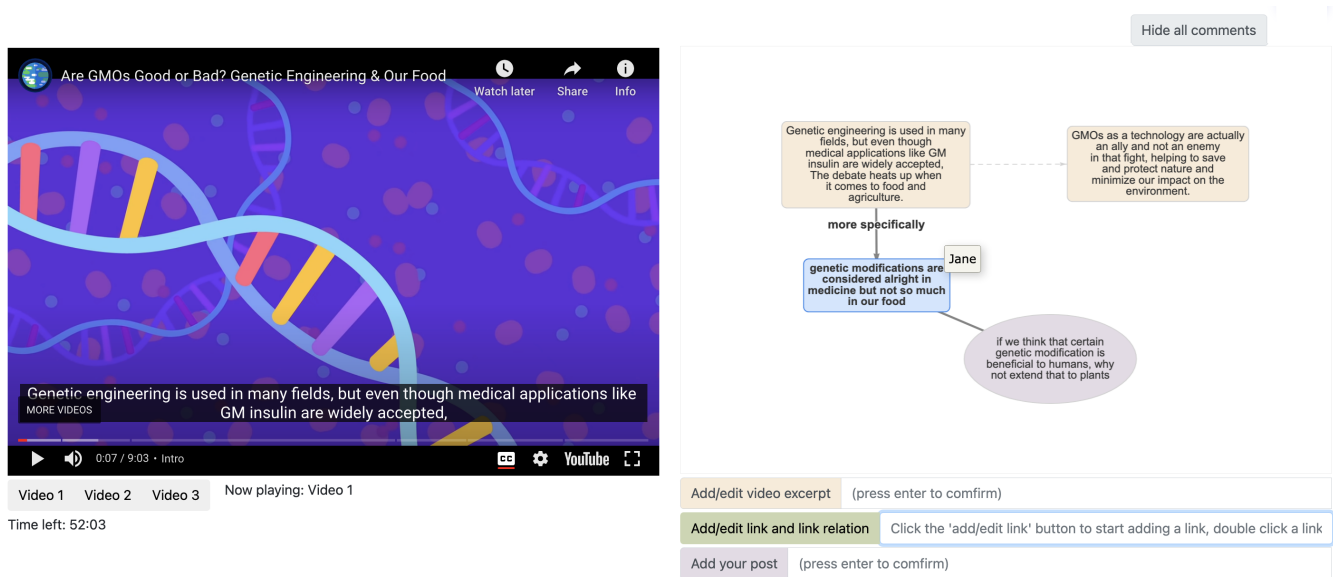


Figure 1: DeepThinkingMap. DeepThinkingMap consists of a video player and a graph-based canvas in parallel. Users may play the video, take notes, and communicate simultaneously as a team, with notes and videos connected via video timestamps.

Positioned adjacent to the video player, the collaborative canvas facilitates simultaneous video consumption and note access and editing. Two primary content input boxes are located beneath the canvas: the 'video excerpt' box for critical takeaways in the video and the 'personal post' box for personal comments and reviews. Different shapes and colors distinguish the two types of notes: box-shaped, light-yellow for video excerpts, and violet ellipsoids for personal posts, reflecting their input box colors. Users can freely add, edit, move, and delete text blocks, links, and link labels during video playback. Directed links between video notes guide users through the relationship's flow, while undirected links connecting to purple comments represent associations between video content and personal reviews. A hover-over feature displays the author's name to indicate contribution. Users can connect any note with other notes, such as forming threaded comments for Q&A. A designed button hides or shows violet ellipsoid inputs, allowing users to focus on video content or both. In practice, optimal utilization of the two input boxes can be collectively determined by the team to distinguish distinct content categories effectively. Users can freely drag and organize the notes in the canvas space to better visualize the association between the notes and comments. Beyond the features to enable concept mapping and commenting, we also consider facilitating the content reference between DeepThinkingMap canvas and video player. Video is hard to search for particular content and refer to as they are unstructured and multi-modal, which needs extra support such as video scrubbing for thumbnails. To help team members better navigate within videos and understand how the notes are created related to the videos, our system records the timestamp of the video when a note is submitted. Users can jump to the note's corresponding video clip by double-clicking. In addition, we also embedded a Discord channel for user communication and coordination of reflection practices.

DeepThinkingMap as a web application is implemented with Firebase, YouTube Data API, and D3.js, among others [1, 12, 13]. The client side utilizes YouTube Data API for video retrieval and D3.js for canvas display and operations. Firebase database is implemented for simultaneous editing with optimistic locking, ensuring all versions are saved and updated in real time for seamless collaboration experiences.

3 INITIAL STUDY IN ASYNCHRONOUS COLLABORATION

We've conducted a between-subject experiment to understand whether and how seeing others' sharing through DeepThinkingMap on the engagement of reflection of health-related videos, which represents a form of asynchronous collaboration [9]. In the study, we manipulated the availability of sharing from other users to design collaboration condition and control condition. Forty-four participants were recruited to watch a health-related video randomly assigned to them and performed video summarizing and commenting. Twenty participants in collaboration condition were given a random average-quality work of notes and comments created by a peer user when they performed the task. And the other twenty-four participants were recruited before the administration of the collaboration condition, and finished the task with an empty canvas. We selected four works among these participants whose comments are of average amount, and used them as peer content that was randomly assigned to participants in the collaboration condition. After the experimental session, we surveyed participants' levels of thinking engagement with modified questionnaires from Kimber et al. [8], including engagement in understanding and reflection. From the data analysis, we found that seeing the other's thoughts and comments improved the engagement in understanding and reflection with the DeepThinkingMap [9]. The finding suggests

that collaboration and exchange of understanding and personal thoughts with DeepThinkingMap may facilitate individuals' understanding and reflection on online videos, which may serve as an effective design strategy for collaborative reflection and resolving misinformation-related issues. Groups may have a more reflective collaboration and transparent exchange with graph-based sharing space design.

4 CONCLUSION AND DEMONSTRATION PLAN

Reflection for high-stake health information is an important yet cognitively taxing and complicated process for the general public. Drawing inspiration from social nudging and concept mapping, we posit that collaborative reflection by exchanging with group thinking activities via shared concept mapping has the potential to help individuals to engage in deeper reflection processes on high-stakes information. In this demo, we introduce DeepThinkingMap, a collaborative reflection system that affords simultaneous video watching and graph-based reflective sharing including content summary and personal comments. The initial study holds promise in engagement promotion for understanding and reflection.

For the demonstration session, we have created multiple sets of videos and collaborative reflection content at varying levels of completion. We will prepare different scenarios that attendees may role-play, showcasing the system's features and approach to facilitate collaborative reflection among teams.

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